LEARNOVA / BIOLOGY

Questions worth understanding.

Parts 1, 2 and 9

Original question images from Zama and Abdulkarim, collected by part. Diagrams, symbols, type styles and original choices are retained.

Part 1 / 51 usable source occurrences

Part 2 / 71 usable source occurrences

Part 9 / 145 usable source occurrences

267 question occurrences. Both source occurrences are retained when questions overlap. This is a focused collection, not a full-book compilation.

How to use this collection

Choose an answer before checking the appendix. Source references use physical PDF page numbers. Original numbering is independent of the course sequence. Worked examples and Try It exercises have separate numbering series.

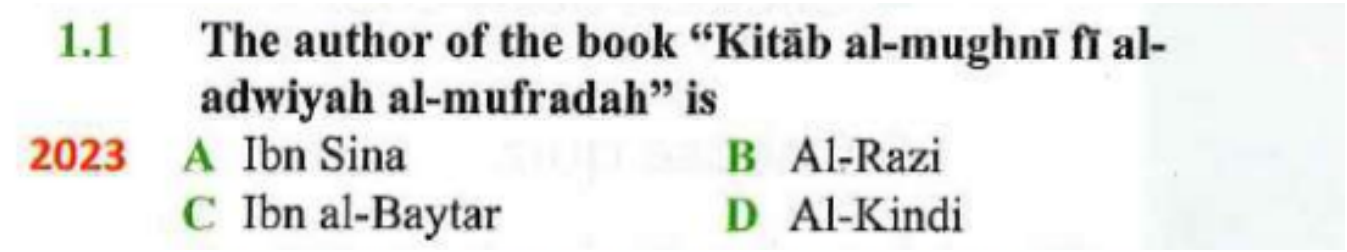
Some source questions have usable images but unresolved answers. Those appendix entries say "Unresolved". Questions with incomplete visible material are omitted from this PDF and flagged in the website. The images preserve original wording, including source spelling and punctuation.

Part 1

Introduction to Biology / source-specific classification material

Zama / Worked examples

Part 1 | Zama Worked 1.1 | PDF page 4 | Printed page 140

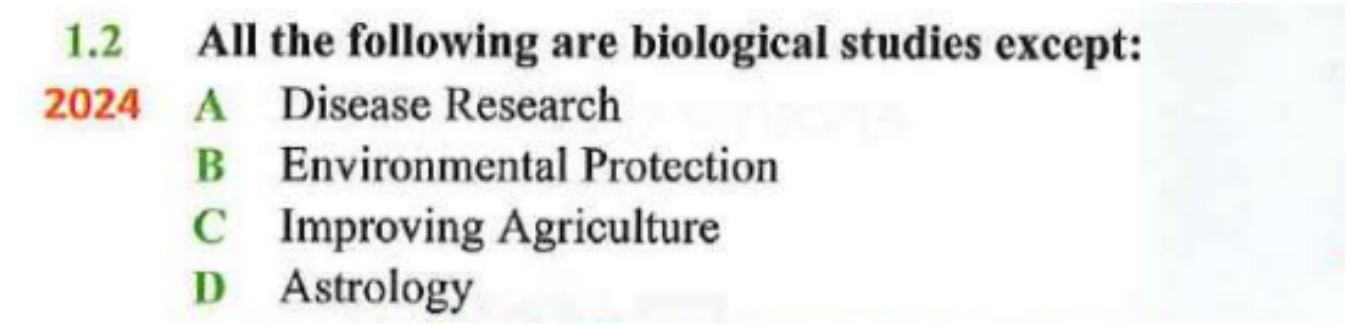


1.1 **The author of the book "Kitāb al-mughnī fī al-adwiyah al-mufradah" is**

2023

A Ibn Sina
B Al-Razi
C Ibn al-Baytar
D Al-Kindi

Part 1 | Zama Worked 1.2 | PDF page 4 | Printed page 140

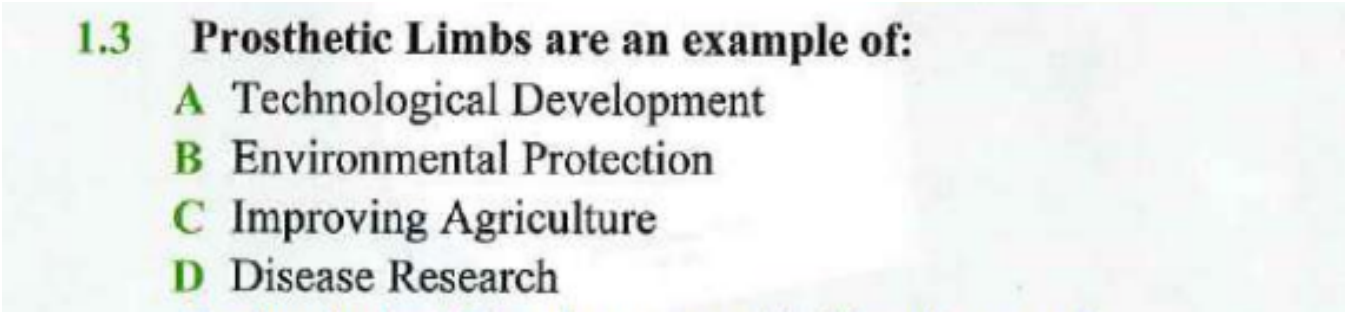


1.2 **All the following are biological studies except:**

2024

A Disease Research
B Environmental Protection
C Improving Agriculture
D Astrology

Part 1 | Zama Worked 1.3 | PDF page 4 | Printed page 140



1.3 **Prosthetic Limbs are an example of:**

A Technological Development
B Environmental Protection
C Improving Agriculture
D Disease Research

Part 1 | Zama Worked 1.4 | PDF page 4 | Printed page 140

1.4 **A researcher studied the genetic makeup of plants, and their ability to resist diseases & insect infestations. This researcher is working on:**

2024
A Technological Development
B Environmental Protection
C Improving Agriculture
D Disease Research

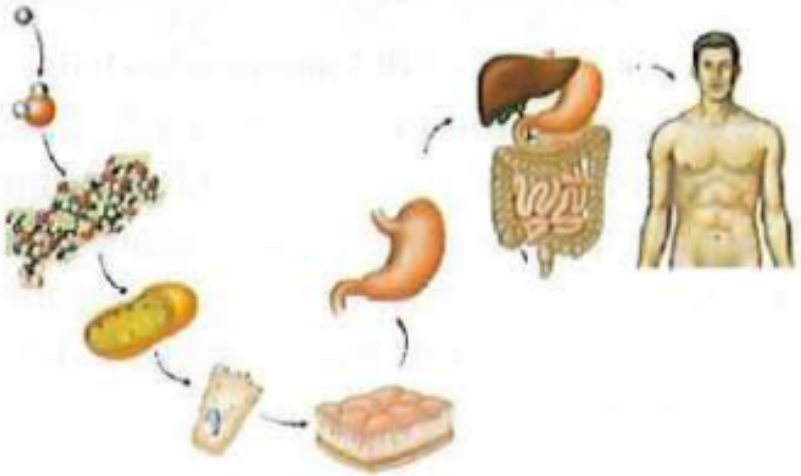
Part 1 | Zama Worked 1.5 | PDF page 5 | Printed page 141

1.5 **The reaction of a living organism to a stimulus is:**

2024
A Response
B Growth
C Adaptation
D Extinction

Part 1 | Zama Worked 1.6 | PDF page 5 | Printed page 141

1.6 **A teacher was explaining the characteristics of living organisms to his students. What characteristic is seen in the following image**



2024
A Organization
B Homeostasis
C Adaptation
D Reproduction

Part 1 | Zama Worked 1.7 | PDF page 5 | Printed page 141

1.7 Some plants live in nitrogen deficient areas. Which of the following methods is used by the plant in order to obtain nitrogen

2024

A Consuming insects
B Symbiotic relationship with bacteria
C Feeding on dead organisms
D Parasitism

Part 1 | Zama Worked 1.8 | PDF page 5 | Printed page 141

1.8 Which of the following correctly describes an organism's ability to withstand a change in environmental conditions

A Growth
B Homeostasis
C Response
D Adaptation

Part 1 | Zama Worked 1.9 | PDF page 5 | Printed page 141

1.9 Maintenance of an organism's internal conditions, essential for survival:

2024

A Adaptation
B Response
C Homeostasis
D Growth

Part 1 | Zama Worked 1.10 | PDF page 5 | Printed page 141

1.10 The basic functional unit of a living organism is

A Tissue
B Cell
C Organ System
D Organ

Part 1 | Zama Worked 1.11 | PDF page 6 | Printed page 142

1.11 As you were going through a group of plants, an odd-looking plant caught your attention. Which of the following steps would come first if you were to conduct a scientific study

2023

A Conclusion
B Observation
C Hypothesis
D Experiment

Part 1 | Zama Worked 1.12 | PDF page 6 | Printed page 142

1.12 In scientific research, which step follows after a scientist noticed a new phenomenon in nature

2023

A Hypothesis
B Conclusion
C Asking a question
D Data Analysis

Part 1 | Zama Worked 1.13 | PDF page 6 | Printed page 142

1.13 A hypothesis is said to be true or false after it has been

2023

A Designed
B Tested
C Analyzed
D Concluded

Part 1 | Zama Worked 1.14 | PDF page 6 | Printed page 142

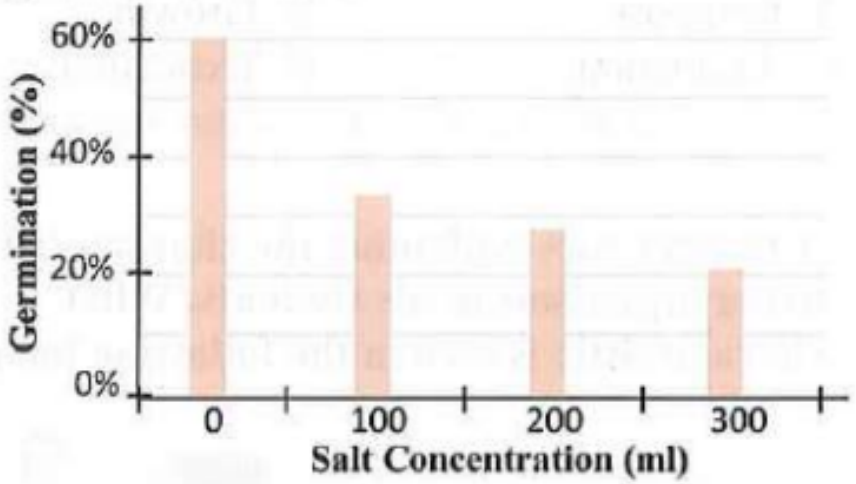
1.14 If a scientist hypothesized that "As light intensity increases, the rate of photosynthesis increases". The scientific method used to collect data in a controlled environment is called

2023

A Experiment
B Conclusion
C Observation
D Analysis

Part 1 | Zama Worked 1.15 | PDF page 6 | Printed page 142

1.15 **The Graph shows the effect of salt concentration on germination. Which of the following is the dependent variable**



2023 A Salt Concentration B Time Taken
C Type of Seeds D Germination

Part 1 | Zama Worked 1.16 | PDF page 6 | Printed page 142

1.16 **After a biologist spent some time observing bats, he established that bats are mammals, this is called a(n):**

2023 A Observation
B Hypothesis
C Analysis
D Conclusion

Part 1 | Zama Worked 1.17 | PDF page 6 | Printed page 142

1.17 **A scientist was conducting a study on a panda. Which of the following units of measurement is used to describe the panda quantitatively**

2023 A Kilogram B Litre
C Mole D Inch

Zama / Try It exercises

Part 1 | Zama Try It 1.1 | PDF page 5 | Printed page 141

1.1 An artificial heart valve comes under the scope of:

Try It?

A Environmental Protection
B Disease Research
C Agricultural Improvement
D Technological Development

Part 1 | Zama Try It 1.2 | PDF page 5 | Printed page 141

1.2 Which of the following would be a major focus for a biologist:

Try It?

A The composition of water in hot springs
B The location of the hot springs.
C Bacteria found in hot springs
D The temperature of hot springs

Part 1 | Zama Try It 1.3 | PDF page 5 | Printed page 141

1.3 Which would be an activity conducted by a biologist?

Try It?

A Design a new, solar-powered car
B Describe the behavior of brown bats
C Identify the cause of a volcanic eruption
D Teach people how to plant corn

Part 1 | Zama Try It 1.4 | PDF page 5 | Printed page 141

1.4 _____ is the study of the origin & history of life, as well as the structure of living organisms:

Try It?

A Physics B Biology C Geology D Zoology

Part 1 | Zama Try It 1.5 | PDF page 5 | Printed page 141

1.5 Try It? **The white hair of a polar bear is an example of**

A Adaptation
B Stimulus
C Homeostasis
D Reproduction

Part 1 | Zama Try It 1.6 | PDF page 6 | Printed page 142

1.6 Try It? **Which of the following best describes adaptation**

A Increase in size
B Short term behavioral change
C Physical change
D A change that resulted from an environmental exposure, making an organism more suited to survive within the environment

Part 1 | Zama Try It 1.7 | PDF page 6 | Printed page 142

1.7 Try It? **The leaves of desert plants adapt to lack of water by all the following except:**

A Curling of the leaves
B Decrease number of stomata
C Increased surface area of the leaf
D Decreasing surface area of the leaf

Part 1 | Zama Try It 1.8 | PDF page 7 | Printed page 143

1.8 **A biologist used a camera to look at the behavior of rabbits, and how they treat their kittens. Which of the following methods was used to obtain this information**

Try It?
A Theory
B Hypothesis
C Observation
D Conclusion

Part 1 | Zama Try It 1.9 | PDF page 7 | Printed page 143

1.9 **A carefully thought-out explanation for observations of the natural world**

Try It?
A Theory
B Hypothesis
C Environment
D Ecology

Part 1 | Zama Try It 1.10 | PDF page 7 | Printed page 143

1.10 **A proposed answer to a scientific study that is testable is**

Try It?
A Hypothesis
B Conclusion
C Analysis
D Theory

Part 1 | Zama Try It 1.11 | PDF page 7 | Printed page 143

1.11 **Jack is testing the effect of temperature on the rate of enzyme activity, which of the following is the independent variable:**

Try It?
A Temperature
B Type of enzyme
C Concentration of Enzyme
D pH of solution

Part 1 | Zama Try It 1.12 | PDF page 7 | Printed page 143

1.12 Mays is testing the effect of nitrate concentration on rate of growth of a plant. Which of the following is the dependent variable:

Try It?

A Light intensity
B Nitrate concentration
C Growth of plant



Abdulkarim / Questions

Part 1 | Abdulkarim Q01 | PDF page 3

01/1 ➢ **Which of the following is not a study of biology…**

a) Environmental protection.
b) Researching diseases.
c) Astronomy.
d) Studying species.

Part 1 | Abdulkarim Q02 | PDF page 3

02/1 ➢ **Prosthetic hands are an example of …**

a) Agricultural improvement.
b) Technology improvement.
c) Preserving the environment.
d) Researching diseases.

Part 1 | Abdulkarim Q04 | PDF page 3

$\frac{04}{1}$

➢ **A group of organisms that can mate and produce offspring that are fertile…**

a) Species.
b) Genus.
c) family
d) Order.

Part 1 | Abdulkarim Q06 | PDF page 3

$\frac{06}{1}$

➢ **Which of the following induces a response in a living organism…**

a) Accumulation.
b) Response.
c) Adaptation.
d) Stimulus.

Part 1 | Abdulkarim Q07 | PDF page 3

$\frac{07}{1}$

➢ **An organism reaction to stimuli…**

a) Response.
b) Growth.
c) Adaptation.
d) Balance.

Part 1 | Abdulkarim Q09 | PDF page 3

$\frac{09}{1}$

➢ **What is the correct scientific name of E.coli…**

a) Escherichia coli.
b) ESCHERICHIA COLI.
c) escherichia coli.
d) Escherichia Coli.

Part 1 | Abdulkarim Q11 | PDF page 4

$\frac{11}{1}$

➢ **Fleming thought that the penicillin secrets a substance that kills bacteria…**

a) Observation.
b) Conclusion.
c) Hypothesis.
d) Law.

Part 1 | Abdulkarim Q12 | PDF page 4

$\frac{12}{1}$

➢ **A researcher observed a bat, and after thinking deeply, he concluded that bats belong to mammals, this is called…**

a) Observation.
b) Analysis.
c) Conclusion.
d) Hypothesis.

Part 1 | Abdulkarim Q13 | PDF page 4

$\frac{13}{1}$

➢ **A unit of measurement in the metric system that can be used to describe the mass of dolphins…**

a) Second.
b) Kilograms.
c) Meter.
d) Liter.

Part 1 | Abdulkarim Q14 | PDF page 4

$\frac{14}{1}$

➢ **Linnaeus classified living organisms based on..**

a) Secondary characteristics.
b) Genetic relations.
c) Binomial nomenclature.
d) Shape (morphology) and behavior.

Part 1 | Abdulkarim Q15 | PDF page 4

$\frac{15}{1}$

In Binomial nomenclature system, the first name Is the name of..

a) Genus.
b) Species.
c) Order.
d) Family.

Part 1 | Abdulkarim Q16 | PDF page 4

$\frac{16}{1}$

Binomial nomenclature gives every organism A scientific name composed of two parts, and they are...

a) Genus and Species.
b) Family and order.
c) Kingdom and phylum.
d) Genus and class.

Part 1 | Abdulkarim Q17 | PDF page 4

$\frac{17}{1}$

What is the correct scientific name of oranges..

a) Citrus Sinensis.
b) citrus sinensis.
c) Citrus sinensis.
d) *citrus Sinensis.*

Part 1 | Abdulkarim Q19 | PDF page 5

$\frac{19}{1}$

Which of the following categories contains one kingdom or more...

a) Genus.
b) Phylum.
c) Family.
d) Domain.

Part 1 | Abdulkarim Q20 | PDF page 5

$\frac{20}{1}$

- **A taxon that contains related classes…**

a) Order. b) Classes.
c) Division. d) Genus.

Part 1 | Abdulkarim Q21 | PDF page 5

$\frac{21}{1}$

- **The term division is used instead of …….. to to classify plants.**

a) Family. b) Order.
c) Class. d) Phylum.

Part 1 | Abdulkarim Q22 | PDF page 5

$\frac{22}{1}$

- **Which of the following texa contains related Families…**

a) Phylum. b) Class.
c) Order. d) Genus.

Part 1 | Abdulkarim Q23 | PDF page 5

$\frac{23}{1}$

- **A group of species that are closely related and share common characteristics and a common ancestor.**

a) Species. b) Genus.
c) Class. d) Phylum.

Part 1 | Abdulkarim Q24 | PDF page 5

24/1 ➢ **Which of the following represents the base unit for classification…**

a) Species. b) Phylum.
c) Family. d) Genus.

Part 1 | Abdulkarim Q25 | PDF page 5

25/1 ➢ **In the modern classification for living organisms, the number of domains…**

a) Three. b) Four.
c) Five. d) Six.

Part 1 | Abdulkarim Q26 | PDF page 5

26/1 ➢ **The modern classification divides living organisms into six…**

a) Classes. b) Phyla (phylum).
c) Kingdoms. d) Domains.

Part 1 | Abdulkarim Q27 | PDF page 5

27/1 ➢ **You have a mushroom: you can classify it within domain…**

a) Fungi.. b) Archaea.
c) Bacteria. d) Eukaryote.

232

Part 2

Classification and microorganisms

Zama / Worked examples

Part 2 | Zama Worked 2.1 | PDF page 8 | Printed page 144

2.1 **The Binomial naming system is made up of 2 names in order, which are:**

2023

A Phylum, Species
B Genus, Species
C Species, Genus
D Classification, Genus

Part 2 | Zama Worked 2.2 | PDF page 8 | Printed page 144

2.2 **Carolus Linnaeus classified living organisms based on:**

2024

A Size & behavior
B Similar Characteristics & Reproduction
C Behavior & Genetic Makeup
D Morphology & Behavior

Part 2 | Zama Worked 2.3 | PDF page 8 | Printed page 144

2.3 **In scientific references, a corn plant is called "Zea mays". This name is made up of**

2023

A Genus , Order
B Order , Phylum
C Genus , Species
D Family , Species

Part 2 | Zama Worked 2.4 | PDF page 8 | Printed page 144

2.4 What is the correct name given to a Tiger based on the binomial naming system:

A *panthera tigris*
B Panthera Tigris
C *Panthera tigris*
D *panthera Tigris*

Part 2 | Zama Worked 2.5 | PDF page 8 | Printed page 144

2.5 Which of the following represents the name given to Escherichia Coli by the binomial naming system

2023

A Escherichia coli
B ESCHERICHIA coli
C escherichia coli
D Escherichia Coli

Part 2 | Zama Worked 2.6 | PDF page 8 | Printed page 144

2.6 The scientific name given to house cats based on the binomial naming system is

2024

A Felis Catus
B felis Catus
C felis catus
D Felis catus

Part 2 | Zama Worked 2.7 | PDF page 9 | Printed page 145

2.7 Which of the following groups is most diverse with regards to the organisms it includes

2023

A Phylum
B Class
C Order
D Family

Part 2 | Zama Worked 2.8 | PDF page 9 | Printed page 145

2.8 **Related organisms that share common characteristics and can produce a fertile offspring:**

A Order
B Species
C Family
D Class

Part 2 | Zama Worked 2.9 | PDF page 9 | Printed page 145

2.9 **The largest level in taxonomic classification is:**

A Domain
B Genus
C Order
D Species

Part 2 | Zama Worked 2.10 | PDF page 10 | Printed page 146

2.10 **Which of the following lack the characteristics of living organisms**

2023

A Bacteria
B Fungi
C Plants
D Viruses

Part 2 | Zama Worked 2.11 | PDF page 10 | Printed page 146

2.11 **The causative agent of AIDS is classified into which of the following**

2023

A A classification of its own because it is nonliving
B The animal kingdom because it is surrounded by a protein
C The bacteria kingdom because it contains genetic material
D With Prokaryotes that withstand harsh conditions

Part 2 | Zama Worked 2.12 | PDF page 10 | Printed page 146

2.12 **How many domains are living organisms classified into**

2024

A 2 B 6 C 5 D 3

Part 2 | Zama Worked 2.13 | PDF page 10 | Printed page 146

2.13 **Modern Taxonomy classifies living organisms into how many kingdoms**

A 2 B 6 C 5 D 3

Part 2 | Zama Worked 2.14 | PDF page 12 | Printed page 148

2.14 **Which of the following organelles are found in bacteria**

2023

A Golgi Apparatus B Ribosomes
C Chloroplasts D Mitochondria

Part 2 | Zama Worked 2.15 | PDF page 12 | Printed page 148

2.15 **Which of the following structures cannot be seen in a bacterial cell under the microscope**

2024

A Cytoplasm B Nuclear Membrane
C Flagellum D DNA

Part 2 | Zama Worked 2.16 | PDF page 12 | Printed page 148

2.16 **Fida discovered a Microorganism near a hot spring which had a temperature of 80°C. What type of Microorganism is it**

2023

A Archaebacteria
B Methanogens
C Streptococcus
D Cyanobacteria

Part 2 | Zama Worked 2.17 | PDF page 12 | Printed page 148

2.17 **Which of the following characteristics is true regarding methanogens**

2023

A Used in sewage treatment
B Respires in presence of oxygen
C They possess a nuclear membrane
D They can photosynthesize

Part 2 | Zama Worked 2.18 | PDF page 12 | Printed page 148

2.18 **When testing sewage, what type of archaea do we expect to see**

2023

A Acidophiles
B Methanogens
C Thermophiles
D Halophiles

Part 2 | Zama Worked 2.19 | PDF page 12 | Printed page 148

2.19 **If gram stain was used to stain a bacterial cell with a thick layer of peptidoglycan, what colour would you expect to see**

2023 **A** Purple **B** Green **C** Pink **D** Orange

Part 2 | Zama Worked 2.20 | PDF page 12 | Printed page 148

2.20 **Most prokaryotic organisms reproduce by which of the following**

2023

A Binary Fission **B** Conjugation
C Budding **D** Fragmentation

Part 2 | Zama Worked 2.21 | PDF page 12 | Printed page 148

2.21 **To identify what type of antibiotic to use, what part of a bacterial cell should be examined**

2023
A Cell Wall
B Ribosomes
C Cell Membrane
D Mitochondria

Part 2 | Zama Worked 2.22 | PDF page 12 | Printed page 148

2.22 **Supposing that a bacterium (Salmonella) fell into a plate of uncovered food, that provided the bacterium with optimum conditions for growth & reproduction. How many bacterial cells do we expect to find after 2 hours if the bacterium takes 20 minutes to replicate**

2023 A 16 B 32 C 128 D 64

Part 2 | Zama Worked 2.23 | PDF page 12 | Printed page 148

2.23 **How do Fava Bean plants benefit from the growth & presence of bacteria on its roots**

2023
A Increased water absorption
B Absorption of minerals
C Provides plant with nitrogen
D Anchor the plant to the ground

Part 2 | Zama Worked 2.24 | PDF page 12 | Printed page 148

2.24 **Escherichia Coli was found in a water well on a farm. They most likely originated from**

2023
A Sewage water B Acid rain
C Medical wastes D Rain

Part 2 | Zama Worked 2.25 | PDF page 12 | Printed page 148

2.25 **A bacterial infection that affects the lungs ability to exchange gases is**

2024

A Lung Cancer
B Asthma
C Influenza
D Tuberculosis

Part 2 | Zama Worked 2.26 | PDF page 12 | Printed page 148

2.26 **When looking at a bacterial cell under the microscope, we expect to see which of the following**

A Nucleus
B Flagellum
C Mitochondria
D Chloroplast

Part 2 | Zama Worked 2.27 | PDF page 13 | Printed page 149

2.27 **Saad isolated a pathogen, upon investigation he found that it was made up of a protein coat that surrounded genetic material. This pathogen is most likely a**

2023

A Virus
B Archaea
C Bacteria
D Fungi

Part 2 | Zama Worked 2.28 | PDF page 13 | Printed page 149

2.28 **Which of the following is correct regarding the contents of all viruses**

2023

A Protein Coat + Cell Membrane
B Protein Capsule + Nucleus + Genetic Material
C Protein Coat + Ribosomes + Nucleus
D Protein Coat + Genetic Material

Part 2 | Zama Worked 2.29 | PDF page 13 | Printed page 149

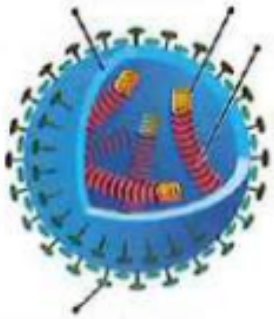
2.29 **The Corona virus is made up of**

2023

A Mitochondria
B Cytoplasm
C Genetic Material
D Lysosomes

Part 2 | Zama Worked 2.30 | PDF page 13 | Printed page 149

2.30 **The structure in the image**
2024 **represents what virus**



A Influenza Virus
B Adenovirus
C Bacteriophage
D Tobacco Mosaic Virus

Part 2 | Zama Worked 2.31 | PDF page 13 | Printed page 149

2.31 **A student was asked to conduct a study on the causative agent of chickenpox. Which of the following scientific topics will help this student conduct his study**

2023

A Viral Diseases
B Bacterial Diseases
C Parasitic Worms
D Diseases Transmitted by Insect Vectors

Part 2 | Zama Worked 2.32 | PDF page 14 | Printed page 150

2.32 **Which of the following statements is incorrect regarding viruses**

2023

A Contains genetic material
B Possesses a protein coat
C Can cause disease
D Viral infections are treated with antibiotics

Part 2 | Zama Worked 2.33 | PDF page 14 | Printed page 150

2.33 **HIV is classified as a**

A Prion
B Retrovirus
C Coxsackievirus
D Adenovirus

Part 2 | Zama Worked 2.34 | PDF page 14 | Printed page 150

2.34 **Viruses are classified based on**

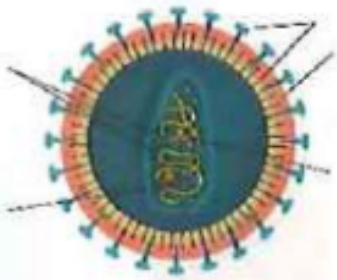
2023

A Size
B Protein Coat
C Genetic Material
D Disease it causes

Part 2 | Zama Worked 2.35 | PDF page 14 | Printed page 150

2.35 **The image shows a retrovirus that causes**

2024



A Chickenpox
B AIDS
C Influenza
D Common Cold

Part 2 | Zama Worked 2.36 | PDF page 14 | Printed page 150

2.36 **Which of the following can infect neurones in the brain**

2023

A Streptococcus
B Prions
C AIDS
D Influenza Virus

Part 2 | Zama Worked 2.37 | PDF page 14 | Printed page 150

2.37 **Which of the following is a sexually transmitted disease**

A Genital Herpes
B Polio Myelitis
C Measles
D Mumps

Zama / Try It exercises

Part 2 | Zama Try It 2.1 | PDF page 8 | Printed page 144

2.1 **What is the correct name given to a human based on the Binomial naming system**

Try It?

A *Homo Sapiens*
B *Homo sapiens*
C *HOMO SAPIENS*
D *homo Sapiens*

Part 2 | Zama Try It 2.2 | PDF page 8 | Printed page 144

2.2 **The organization of living organisms into groups, based on similarities in their characteristics and evolutionary basis is:**

Try It?

A Classification
B Adaptation
C Species
D Orientation

Part 2 | Zama Try It 2.3 | PDF page 8 | Printed page 144

2.3 2024 Try It?

What is the scientific name given to an Orange when printed in a book:

A Citrus *Sinensis* B *citrus Sinensis*
C *Citrus sinensis* D *citrus sinensis*

Part 2 | Zama Try It 2.4 | PDF page 8 | Printed page 144

2.4 Try It?

What is the scientific name given to an Apple tree when written on a paper:

A Malus Domestica B Malus domestica
C malus domestica D Malus domestica

Part 2 | Zama Try It 2.5 | PDF page 9 | Printed page 145

2.5 Try It?

The smallest level in taxonomic classification

A Domain B Genus C Species D Order

Part 2 | Zama Try It 2.6 | PDF page 9 | Printed page 145

2.6 2024 Try It?

Which taxonomic level comes in third place (descending order):

A Phylum B Genus C Class D Order

Part 2 | Zama Try It 2.7 | PDF page 9 | Printed page 145

2.7 Try It?

Which taxonomic level comes in fourth place (ascending order):

A Genus B Family C Order D Species

Part 2 | Zama Try It 2.8 | PDF page 10 | Printed page 146

2.8 Which of the following kingdoms does not fall under the domain Eukaryota

Try It?

A Animalia
B Archaebacteria
C Protists
D Fungi

Part 2 | Zama Try It 2.9 | PDF page 10 | Printed page 146

2.9 All the following regarding Archaea are correct except

Try It?

A Includes the kingdom Archaebacteria
B Can survive in harsh conditions
C Absence of membrane bound organelles
D Cell wall contains peptidoglycan

Part 2 | Zama Try It 2.10 | PDF page 10 | Printed page 146

2.10 Which of the following have membrane bound organelles

Try It?

A Animalia
B Archaebacteria
C Viruses
D Bacteria

Part 2 | Zama Try It 2.11 | PDF page 10 | Printed page 146

2.11 What are plant cell walls made up of

Try It?

A Peptidoglycan
B Chitin
C Cellulose
D Protein

Part 2 | Zama Try It 2.12 | PDF page 10 | Printed page 146

2.12 **Members of the Protist kingdom were classified into 3 groups based on**

Try It?

A Movement
B Mode of nutrition
C Absence of membrane bound organelles
D Modes of reproduction

Part 2 | Zama Try It 2.13 | PDF page 12 | Printed page 148

2.13 **What colour would the cell wall of a bacteria stain if it had a thin layer of peptidoglycan, & high lipid content**

Try It?

A Red B Green C Purple D Pink

Part 2 | Zama Try It 2.14 | PDF page 13 | Printed page 149

2.14 **Which of the following is not a respiratory disease**

Try It?

A Pneumonia
B Cholera
C Whooping Cough
D Tuberculosis

Part 2 | Zama Try It 2.15 | PDF page 13 | Printed page 149

2.15 **Vitamin K is produced by which of the following bacteria**

Try It?

A Escherichia coli
B Acne Vulgaris
C Staphylococcus Aureus
D Mycobacterium Tuberculosis

Part 2 | Zama Try It 2.16 | PDF page 14 | Printed page 150

2.16 Try It?

Which of the following is a disease caused by prions

A Rabies
B Mad Cow Disease
C Pneumonia
D Influenza

Part 2 | Zama Try It 2.17 | PDF page 14 | Printed page 150

2.17 Try It? 2023

Which of the following results due to prions infecting nerve cells in the brain

A Creutzfeldt-Jakob disease
B Mumps
C Pneumonia
D Polio

Part 2 | Zama Try It 2.18 | PDF page 14 | Printed page 150

2.18 Try It?

Which of the following is incorrect regarding viruses

A Protein coat contains genome
B Can cause infection
C Influenza Virus causes respiratory disease
D Contain a Nucleus

Abdulkarim / Questions

Part 2 | Abdulkarim Q01 | PDF page 7

$\frac{01}{2}$

- **A scientist discovered a new organism and concluded it was prokaryote based on what?**

a) Small vacuoles
b) There is ribosome in their cytoplasm
c) Presence of cell wall
d) Lack of membrane-bond organelles

Part 2 | Abdulkarim Q02 | PDF page 7

$\frac{02}{2}$

- **Prokaryote use flagella for …**

a) Attach to surface
b) Prevent the cell from drying out
c) feeding
d) Movement

Part 2 | Abdulkarim Q03 | PDF page 7

$\frac{03}{2}$

- **After examining sewage water what type of prokaryote would you find?**

a) Cyanobacteria
b) Methanogen
c) Acidophilic
d) Halophilies

Part 2 | Abdulkarim Q04 | PDF page 7

$\frac{04}{2}$

- **If the cell wall of the bacteria contain a large amount of peptidoglycan then it would stain which gram of color**

a) Pink
b) Purple
c) Yellow
d) Orange

Part 2 | Abdulkarim Q05 | PDF page 7

05/2

A person was infected by a bacterial disease, what should be examined to prescribe the right medicine

a) Ribosomes
b) Chromosomes
c) Cell wall
d) Plasma membrane

Part 2 | Abdulkarim Q06 | PDF page 7

06/2

The relation between nitrogen fixation bacteria and plant root nodule

a) Mutualism
b) saprobes
c) parasites
d) predation

Part 2 | Abdulkarim Q07 | PDF page 7

07/2

A bacteria that is important for producing vitamin (K)

a) Bacteriophage
b) E.coli
c) Autotrophic bacteria
d) Spirochetes

Part 2 | Abdulkarim Q08 | PDF page 8

08/2

Mohammad was able to isolate a pathogen that its genetic material is surrounded by a protein coat.

a) Bacteria
b) viruses
c) fungus
d) prokaryotes

Part 2 | Abdulkarim Q09 | PDF page 8

$\frac{09}{2}$

➢ **Which of the following materials is found all viruses**

a) Genetic material(DNA) and capsule
b) Nucleus, DNA, capsule
c) Nucleus ,capsule, ribosomes
d) Nucleus, DNA, membrane

Part 2 | Abdulkarim Q10 | PDF page 8

$\frac{10}{2}$

➢ **HIV is classified as which virus?**

a) Amphoteric
b) retrovirus
c) hemolytic
d) lytic

Part 2 | Abdulkarim Q11 | PDF page 8

$\frac{11}{2}$

➢ **Which of the following is a virus disease?**

a) Tuberculosis
b) Cholera
c) Tetanus
d) Aids

Part 2 | Abdulkarim Q12 | PDF page 8

$\frac{12}{2}$

➢ **The influenza virus that replicate by…**

a) Lytic cycle
b) Lysogenic cycle
c) cell cycle
d) organic cycle

Part 2 | Abdulkarim Q13 | PDF page 8

$\frac{13}{2}$ ➢ **The genetic material in a virus that is injected in a chromosome of a host cell through…**

a) Lytic cycle b) Lysogenic cycle
c) Cell cycle d) Organic cycle

Part 2 | Abdulkarim Q14 | PDF page 8

$\frac{14}{2}$ ➢ **Which of the following virus replicate through lysogenic cycle?**

a) Herpes simplex I b) Influenza
c) Common cold d) Tuberculosis

Part 2 | Abdulkarim Q15 | PDF page 8

$\frac{15}{2}$ ➢ **A protein that can cause infection or disease and it's called proteinaceous……**

a) Virus b) Bacteria
c) Prion d) Germs

Part 2 | Abdulkarim Q16 | PDF page 8

$\frac{16}{2}$ ➢ **Which of the following can infect nerve cell in the brain?**

a) Herpes b) Prion
c) Aids d) Tuberculosis

Part 9

Genetics and inheritance

Zama / Worked examples

Part 9 | Zama Worked 9.1 | PDF page 93 | Printed page 229

9.1 2023 **According to Mendel's 1st Law, when crossing a black rabbit (BB) with a white rabbit (bb), the 1st generation will have which of the following genotypes**

A Bb B BB C bb D Bbb

Part 9 | Zama Worked 9.2 | PDF page 93 | Printed page 229

9.2 **When crossing a homozygous dominant red flowered plant, with a homozygous recessive white flowered plant, the 1st generation will be**

2023 A Homozygous & Red B Homozygous & White
C Hybrid & Red D Hybrid & Purple

Part 9 | Zama Worked 9.3 | PDF page 93 | Printed page 229

9.3 **If both parents have a heterozygous genotype (Tt) & the ability to twist their tongues. What genotypes are likely to be seen in their children**

2024 A TT , Tt , tt B TT Only
C tt Only D TT , tt Only

Part 9 | Zama Worked 9.4 | PDF page 93 | Printed page 229

9.4 **The crossing of 2 plants resulted in the formation of a red & a white flowered plant. You expect the parents genotype to be**

2023 A RR & rr B RR & RR
C rr & rr D Rr & Rr

Part 9 | Zama Worked 9.5 | PDF page 93 | Printed page 229

9.5
2023

In fruit flies, the red back trait (R) is considered dominant to the black back trait (r) which is recessive. What is the phenotypic ratio seen when you cross a black back fly with a heterozygous female

A 1 : 1 B 1 : 2 C 2 : 1 D 3 : 1

Part 9 | Zama Worked 9.6 | PDF page 94 | Printed page 230

9.6

The trait seen in the 1st Filial Generation based on Mendel's experiment is

2023

A Dominant B Recessive
C Sex Linked D Polygenic

Part 9 | Zama Worked 9.7 | PDF page 94 | Printed page 230

9.7
2023

A male animal with curly ears bred with a female animal with normal ears. Their children (1st generation) all had normal ears. When the children of the 1st generation reproduced with one another, their phenotypic ratio was 3 normal eared offsprings to 1 curly eared offspring (3:1). The curly ear trait is said to be

A Crossing Over B Linked Genes
C Dominant D Recessive

Part 9 | Zama Worked 9.8 | PDF page 95 | Printed page 231

9.8

♀ / ♂	AB	Ab	aB	ab
AB	AABB	AABb	AaBB	AaBb
Ab	AABb	AAbb	AaBb	Aabb
aB	AaBB	AaBb	aaBB	aaBb
ab	AaBb	Aabb	aaBb	?

The table above is a punnett square, it shows a light-coloured seed (A), & dark coloured seed (a), & round seed (B), & a wrinkled seed (b). What is the phenotype of the genotype in the marked box

2024

A Light round seed B Light wrinkled seed

C Dark round seed D Dark wrinkled seed

Part 9 | Zama Worked 9.9 | PDF page 95 | Printed page 231

9.9

Number	Genotype	Homozygous	Heterozygous
1	RrTT	✓	×
2	RRTT	✓	×
3	RrTt	×	✓
4	rrtt	×	✓

In the table above, which of the following rows is correct regarding the 1st generations genotype that results from a cross between a tall red flowered plant (RRTT), & a short white flowered plant (rrtt)

A 1 B 2 C 3 D 4

Part 9 | Zama Worked 9.10 | PDF page 95 | Printed page 231

9.10 **If an organism's sex cell had 4 pairs of chromosomes, how many genetic combinations do you expect to see after fertilization**

2023 A 8 B 12 C 14 D 16

Part 9 | Zama Worked 9.11 | PDF page 96 | Printed page 232

9.11 **The genetic disease that affects sweat production & makes mucus thicker is called**

2023 A Cystic Fibrosis B Albinism
C Galactosemia D Huntington

Part 9 | Zama Worked 9.12 | PDF page 96 | Printed page 232

9.12 **If 2 parents were carriers of Cystic Fibrosis, an autosomal recessive disorder. What is the probability of them having a diseased child**

2023 A 25% B 50% C 75% D 100%

Part 9 | Zama Worked 9.13 | PDF page 96 | Printed page 232

9.13 **A genetic disorder that results in the absence of melanin pigments in the skin, hair, & eyes is called**

2023 A Cystic Fibrosis B Albinism
C Tay-Sachs Disease D Galactosemia

Part 9 | Zama Worked 9.14 | PDF page 96 | Printed page 232

9.14 **Which of the following is the cause of Albinism**

2023

A A mutation within the gene that codes for a membrane protein

B The absence of the gene that codes for the enzyme responsible for the breakdown of galactose

C Absence of the gene that is responsible for the production of the enzyme that breaks down fatty substances

D Genetic defect that results in the absence of melanin

Part 9 | Zama Worked 9.15 | PDF page 96 | Printed page 232

9.15 **A genetic disorder that results in the accumulation of lipids within the brain**

2023

A Galactosemia B Albinism

C Cystic Fibrosis D Tay-Sachs

Part 9 | Zama Worked 9.16 | PDF page 96 | Printed page 232

9.16 **A genetic disorder that results in the body's inability to digest galactose is called**

A Galactosemia B Tay-Sachs

C Huntington D Cystic Fibrosis

Part 9 | Zama Worked 9.17 | PDF page 96 | Printed page 232

9.17 **Which of the following foods should be avoided in individuals with Galactosemia**

2023

A Chicken B Fish C Yoghurt D Lentils

Part 9 | Zama Worked 9.18 | PDF page 96 | Printed page 232

9.18 **A healthy individual that is heterozygous for a recessive genetic disorder is called**

2023

A Carrier
B Diseased
C Mild Penetrance
D Complete Penetrance

Part 9 | Zama Worked 9.19 | PDF page 97 | Printed page 233

9.19 **In humans, Huntington's disease affects which of the organ systems**

2023

A Nervous System
B Circulatory System
C Respiratory System
D Digestive System

Part 9 | Zama Worked 9.20 | PDF page 97 | Printed page 233

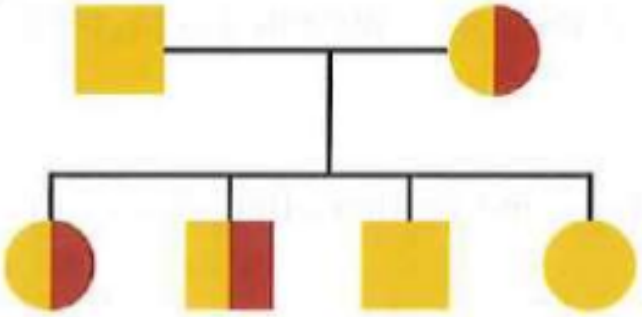
9.20 **If an individual has a small body along with short limbs, & a large head. They most likely have**

2023

A Huntington's Disease
B Achondroplasia
C Albinism
D Galactosemia

Part 9 | Zama Worked 9.21 | PDF page 97 | Printed page 233

9.21 **You are following a certain trait in a pedigree diagram. Which of the following statements is true regarding the individuals of the 2nd generation**



2023

A A male & female show the trait

B It is a sex-linked trait

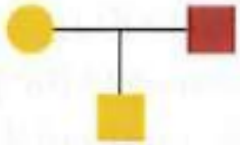
C The gene that results in the trait is only dominant in females

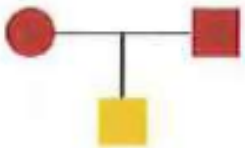
D The ratio or carriers to healthy individuals is equal

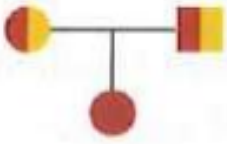
Part 9 | Zama Worked 9.22 | PDF page 97 | Printed page 233

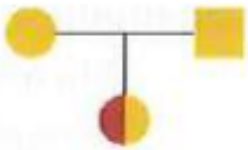
9.22 **Which of the following pedigree diagrams is correct**

2023

A 

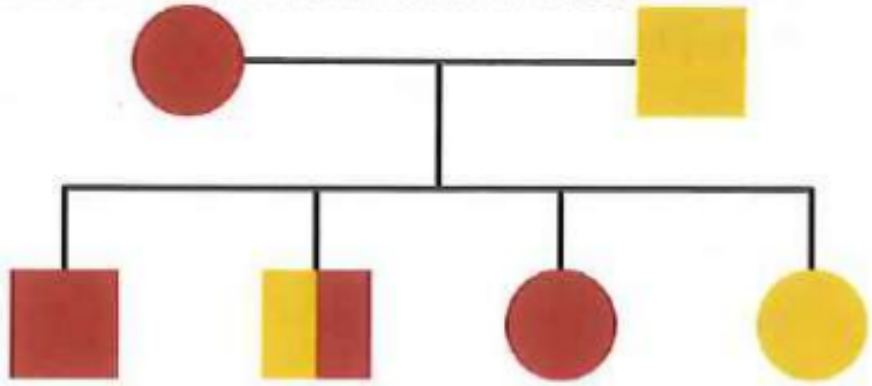
B 

C 

D 

Part 9 | Zama Worked 9.23 | PDF page 97 | Printed page 233

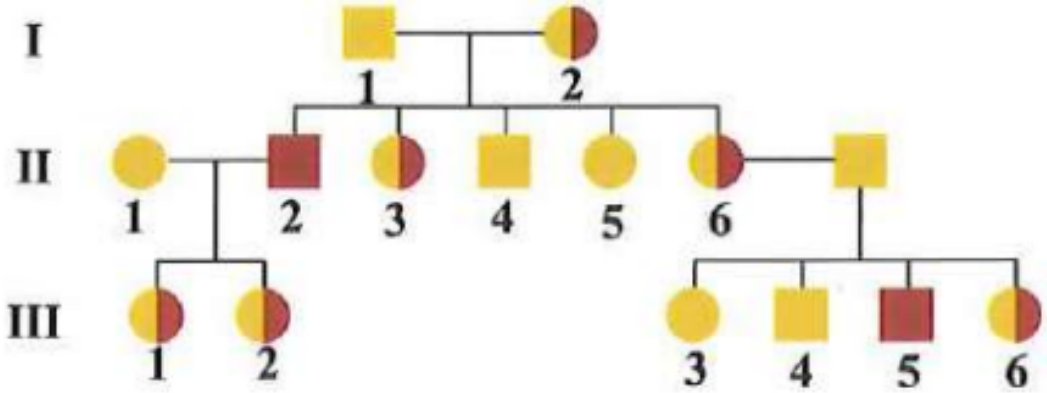
9.23 **In the following pedigree diagram, how many diseased males & females are there**



2024 A 1 Male, 1 Female B 1 Male, 2 Female
C 2 Male, 2 Female D 2 Male, 1 Female

Part 9 | Zama Worked 9.24 | PDF page 98 | Printed page 234

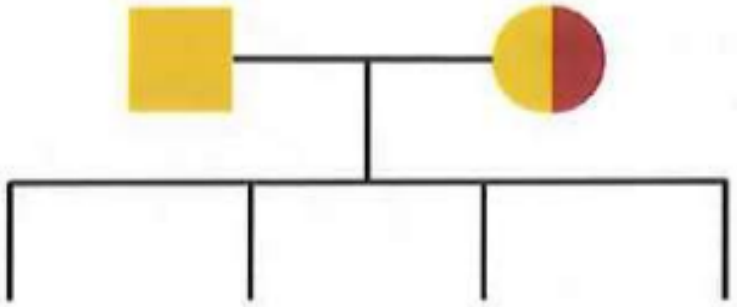
9.24 **In the pedigree diagram, individual II-2 is**



2024 A Carrier Female B Diseased Male
C Healthy Female D Carrier Male

Part 9 | Zama Worked 9.25 | PDF page 98 | Printed page 234

9.25 **The pedigree diagram below shows a certain trait expressed by the parents. Which of the following represents their offsprings genotypes**

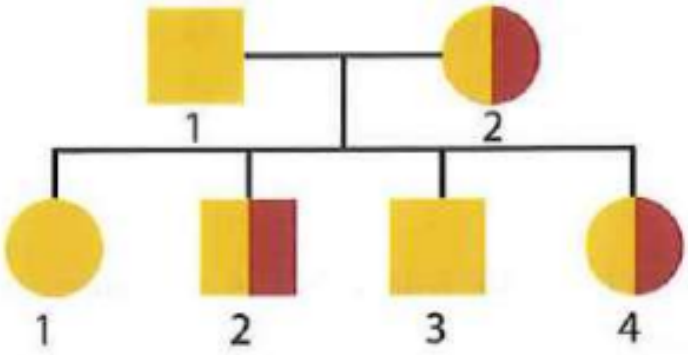


2023 A  B 

C D

Part 9 | Zama Worked 9.26 | PDF page 98 | Printed page 234

9.26 **In the pedigree diagram below, what is the ratio of carriers to normal individuals**



2023 A 1 : 1 B 1 : 2 C 2 : 1 D 3 : 1

Part 9 | Zama Worked 9.27 | PDF page 98 | Printed page 234

9.27 **Using the following punnett square, what is the probability of developing a pink flowered Marvel-of-Peru plant**

	R	r
R	RR	Rr
r	Rr	rr

2024 A 25% B 50% C 75% D 100%

Part 9 | Zama Worked 9.28 | PDF page 99 | Printed page 235

9.28 **The inheritance of A, B, & O blood groups in humans follows with of the following modes**

2023
A Co-Dominance
B Multiple Alleles
C Incomplete Dominance
D Epistasis

Part 9 | Zama Worked 9.29 | PDF page 99 | Printed page 235

9.29 **If a mother has blood group A, & the father has blood group AB. Which of the following blood groups is their offspring least likely to have**

2024 A A B AB C B D O

Part 9 | Zama Worked 9.30 | PDF page 99 | Printed page 235

9.30 **After Jack was born, 4 families disputed over who his parents where. If Jack's has blood group O, which of the following families are least likely to be his parents**

2023
A A Father & B Mother
B AB Father & O Mother
C B Father & O Mother
D O Father & A Mother

Part 9 | Zama Worked 9.31 | PDF page 99 | Printed page 235

9.31 **The genotype of blood group AB is**

2023 A I^AI^A B I^AI^B C I^BI^B D ii

Part 9 | Zama Worked 9.32 | PDF page 99 | Printed page 235

9.32 **The genes I^A & I^B found in blood groups are an example of which of the following**

2023
A Complete Dominance
B Co-Dominance
C Incomplete Dominance
D Mendelian Dominance

Part 9 | Zama Worked 9.33 | PDF page 99 | Printed page 235

9.33 **The most likely genotype of the following rabbit's phenotype is**



A CC B $c^{ch}c$ C c^hc^h D cc

Part 9 | Zama Worked 9.34 | PDF page 100 | Printed page 236

9.34 **If sex cells have 23 chromosomes, how many chromosomes are present in a skin cell**

2023 A 69 B 44 C 46 D 23

Part 9 | Zama Worked 9.35 | PDF page 100 | Printed page 236

9.35 **If sex cells in chickens have 39 chromosomes, how many chromosomes would you expect their liver cells to have**

2023 A 19 B 39 C 78 D 156

Part 9 | Zama Worked 9.36 | PDF page 101 | Printed page 237

9.36 **Traits that are carried on the X chromosome are called**

2023

A Multiple Alleles
B Sex-Linked Traits
C Sex-Influenced Traits
D Recessive Lethal Alleles

Part 9 | Zama Worked 9.37 | PDF page 101 | Printed page 237

9.37 **Which of the following is an X-Linked disorder**

A Short Sightedness B Down Syndrome
C Albinism D Haemophilia

Part 9 | Zama Worked 9.38 | PDF page 101 | Printed page 237

9.38 **A colour blind individual cannot differentiate between which of the following colours**

2023

A Red & Green B Grey & Brown
C Black & White D Yellow & Orange

Part 9 | Zama Worked 9.39 | PDF page 101 | Printed page 237

9.39 **A healthy male married a female carrier of red-green colour blindness. What is the probability of them having a male child with red-green colour blindness**

2023 A 0% B 25% C 50% D 100%

Part 9 | Zama Worked 9.40 | PDF page 101 | Printed page 237

9.40 **Which of the following is a Sex-Influenced trait**

2023

A Skin colour B Colour Blindness
C Bleeding Tendency D Alopecia

Part 9 | Zama Worked 9.41 | PDF page 101 | Printed page 237

9.41 2023 **Alopecia is a Sex-Influenced trait, that is dominant in males & recessive in females. If B represents bald, & b represents not bald. Which of the following genotypes represents a bald female**

A bB B bb C Bb D BB

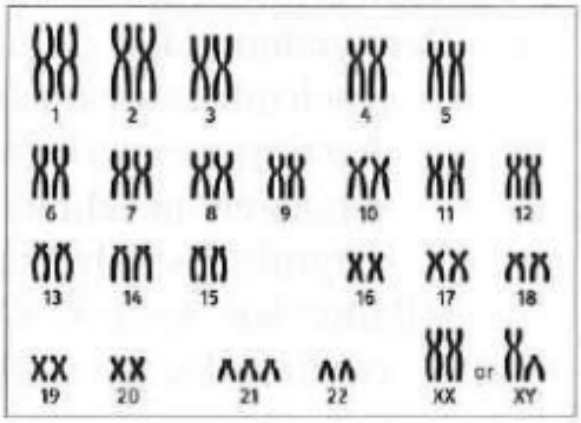
Part 9 | Zama Worked 9.42 | PDF page 101 | Printed page 237

9.42 **Chromosomes need telomeres for**

2023 A Protection whilst crossing over
B Sticking fragments together in order to form DNA
C Formation of tRNA
D Points of connection with spindle fibres

Part 9 | Zama Worked 9.43 | PDF page 101 | Printed page 237

9.43 **The following karyotype represents which of the following diseases**



2024 A Turner Syndrome
B Down Syndrome
C Colour Blindness
D Klinefelter Syndrome

Part 9 | Zama Worked 9.44 | PDF page 102 | Printed page 238

9.44 **Individuals with Turner Syndrome have which of the following genotypes**

2023 A XO B XX C XY D OY

Part 9 | Zama Worked 9.45 | PDF page 102 | Printed page 238

9.45 **Individuals with the genotype XXY have what phenotype**

2024 A Down Syndrome B Klinefelter Syndrome
C Normal Male D Normal female

Part 9 | Zama Worked 9.46 | PDF page 102 | Printed page 238

9.46 **Which of the following genotypes causes spontaneous death**

2023 A OY B OX C XXY D XYY

Part 9 | Zama Worked 9.47 | PDF page 102 | Printed page 238

9.47 **The scientist that analyzed the amount of Adenine, Thymine, Guanine, & Cytosine was**

2023 A Chargaff B Griffith C Hershey D Chase

Part 9 | Zama Worked 9.48 | PDF page 102 | Printed page 238

9.48 **The first to discover DNA as genetic material was**

2023 A Avery B Hershey C Griffith D Chargaff

Part 9 | Zama Worked 9.49 | PDF page 102 | Printed page 238

9.49 **The following image represents**



2024 A DNA B mRNA C rRNA D tRNA

Part 9 | Zama Worked 9.50 | PDF page 102 | Printed page 238

9.50 **Which of the following nitrogenous bases is not found in RNA**

A Cytosine
B Uracil
C Thymine
D Guanine

Part 9 | Zama Worked 9.51 | PDF page 103 | Printed page 239

9.51 **Which of the following statements describes the function of nucleic acids**

2024
A Storage & transfer of genetic material
B Storage of energy & provides structural support
C Storage of energy & barrier formation
D Transport of substances & increase rate of reaction

Part 9 | Zama Worked 9.52 | PDF page 103 | Printed page 239

9.52 **Uracil is found in RNA in place of**

2024
A Adenine
B Guanine
C Cytosine
D Thymine

Part 9 | Zama Worked 9.53 | PDF page 103 | Printed page 239

9.53 **Which of the following represents the correct complementary pairing between nitrogenous bases**

A A – T G – C
B U – C A – G
C A – G T – U
D G – U C – A

Part 9 | Zama Worked 9.54 | PDF page 103 | Printed page 239

9.54 **Regarding the sequence of nitrogenous bases in RNA, which of the following statements is correct**

2023

A Adenine pairs Thymine with 2 hydrogen bonds
B Adenine pairs Thymine with 3 hydrogen bonds
C Adenine pairs Uracil with 2 hydrogen bonds
D Adenine pairs Uracil with 3 hydrogen bonds

Part 9 | Zama Worked 9.55 | PDF page 103 | Printed page 239

9.55 **While studying a molecule of DNA you found that cytosine constitutes 30% of it. According to Chargaff's rule, what percentage of guanine do you expect to see**

2023

A 10% B 20% C 30% D 40%

Part 9 | Zama Worked 9.56 | PDF page 104 | Printed page 240

9.56 **If in a nucleic acid the sequence of bases was 5' TGAAGTTA 3'. What is the sequence seen in the complementary strand**

2023

A 3' ACTTCAAT 5' B 5' ACTTCAA 3'
C 3' CAGGACCG 5' D 5' CAGGACCG 3'

Part 9 | Zama Worked 9.57 | PDF page 104 | Printed page 240

9.57 **The sequence 3' AGATTCGA 5' is found on a strand of DNA. Which of the following sequences is complementary to this strand**

A 5' UCUAAGCU 3' B 3' TCTAAGCT 5'
C 3' ACAUUGCA D 5' TCTAAGCT 3'

Part 9 | Zama Worked 9.58 | PDF page 104 | Printed page 240

9.58 **The complementary strand to the following sequence 5' ATGGGCGC 3' is**

2024 A 3' TAGGGCGG 5' B 3' ATCGGCCG 5'
C 3' TACCCGCG 5' D 3' TAGCGCGG 5'

Part 9 | Zama Worked 9.59 | PDF page 104 | Printed page 240

9.59 **What type of RNA carries amino acids that are free in the cytoplasm to ribosomes**

2023 A tRNA B mRNA C rRNA D cRNA

Part 9 | Zama Worked 9.60 | PDF page 104 | Printed page 240

9.60 **In order to form a protein made up of 60 amino acids, the number of nucleotides found in mRNA is**

2023 A 60 B 120 C 180 D 240

Part 9 | Zama Worked 9.61 | PDF page 105 | Printed page 241

9.61 **If part of a DNA strand has the following sequence AGCAATTGG. What is the sequence of nucleotides in a complementary mRNA strand**

A UCGUUAACC B TCGTTAACC
C AUCAAUUGG D ATCAATTGG

Part 9 | Zama Worked 9.62 | PDF page 105 | Printed page 241

9.62 **Which of the following is a start codon**

2023 A UAG B UAA C UGA D AUG

Part 9 | Zama Worked 9.63 | PDF page 105 | Printed page 241

9.63 **Which of the following is a stop codon in mRNA**

A AUG B AUU C UAA D CAU

Part 9 | Zama Worked 9.64 | PDF page 105 | Printed page 241

9.64 **The process in which mRNA binds to ribosomes, & proteins are formed is called**

2023

A Transcription
B Translation
C Dephosphorylation
D Duplication

Part 9 | Zama Worked 9.65 | PDF page 106 | Printed page 242

9.65 **The modification of a living organisms DNA by the addition of an external sequence of DNA to the original genome is called**

2024

A Genome
B Genetic Engineering
C Mutation
D Bioremediation

Part 9 | Zama Worked 9.66 | PDF page 106 | Printed page 242

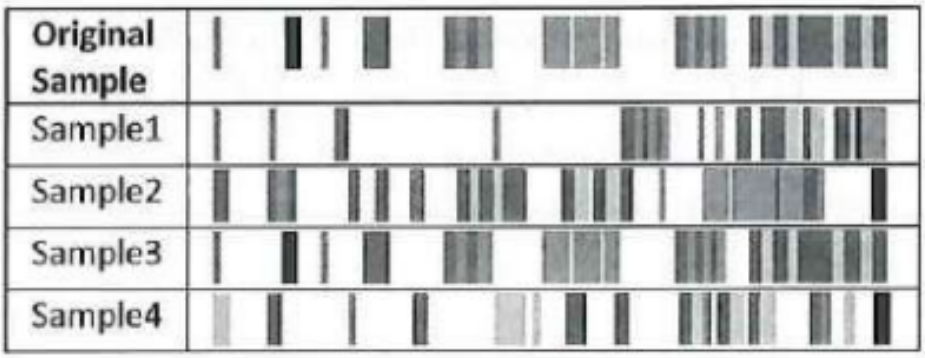
9.66 **The correct statement regarding germline mutations in sex cells is**

2023

A Mutations disappear when the gametes are formed
B Mutations seen in the 1st generation of offsprings
C Can be medically treated or cured
D None of the above

Part 9 | Zama Worked 9.67 | PDF page 106 | Printed page 242

9.67 2023 **A piece of hair was found at a murder scene. It turned out that this piece of hair matched the description of the murderer. In order to confirm this suspicion a sample of DNA was obtained from the hair sample & analyzed. The result was then compared with the other murder suspects. Looking at the DNA analysis below, who is the murderer?**



A 1 B 2 C 3 D 4

Zama / Try It exercises

Part 9 | Zama Try It 9.1 | PDF page 94 | Printed page 230

9.1 **The transfer of a trait from parent to offspring during reproduction is called**

Try It? A Inheritance B Division C Dominance D None of the above

Part 9 | Zama Try It 9.2 | PDF page 94 | Printed page 230

9.2 **Who is the father of genetics**

Try It? A Louis Pasteur B Thomas Edison C Nikola Tesla D Gregor Mendel

Part 9 | Zama Try It 9.3 | PDF page 94 | Printed page 230

9.3 **The genotype seen when crossing a green pea (yy) with a yellow pea (YY) is**

Try It? A YY B yy C Yy D YY , yy

Part 9 | Zama Try It 9.4 | PDF page 95 | Printed page 231

9.4 **If 12 chromosome pairs were seen in an organism's sex cell, how many genetic combinations do you expect there to be after fertilization**

Try It? A 24 B 3578 C 4096 D 5437

Part 9 | Zama Try It 9.5 | PDF page 96 | Printed page 232

9.5 **The mutated gene responsible for the development of Tay-Sachs disease is found on what chromosome**

Try It? A 17 B 21 C 15 D 4

Part 9 | Zama Try It 9.6 | PDF page 96 | Printed page 232

9.6 **Tay-Sachs is most commonly seen in what population of people**

Try It?
A West African
B Southern European Jews
C African American
D Southeast Asian

Part 9 | Zama Try It 9.7 | PDF page 96 | Printed page 232

9.7 How is Tay-Sachs disease diagnosed

Try It?

A Finding a red spot at the back of the eye
B Increased levels of blood galactose
C Recurrent respiratory infections
D It cannot be diagnosed

Part 9 | Zama Try It 9.8 | PDF page 97 | Printed page 233

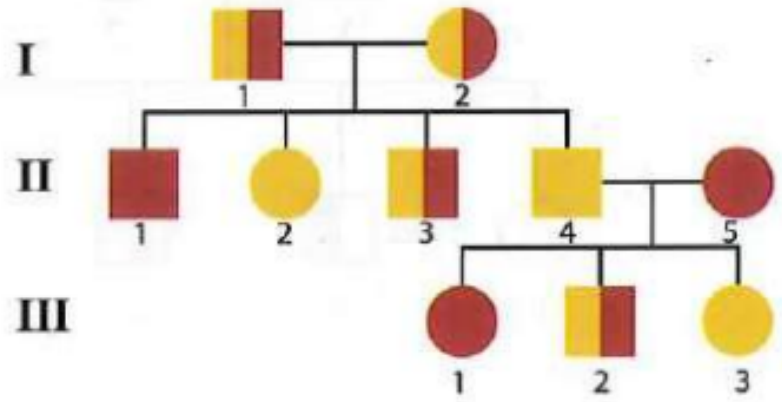
9.8 Which of the following is a dominant genetic disorder

Try It?

A Cystic Fibrosis
B Galactosemia
C Huntington's Disease
D Albinism

Part 9 | Zama Try It 9.9 | PDF page 98 | Printed page 234

9.9 In the pedigree diagram below, which family member is healthy, & has a diseased child



Try It?

A I-1
B I-2
C II-4
D III-2

Part 9 | Zama Try It 9.10 | PDF page 98 | Printed page 234

9.10 **Using the following punnett square, what is the probability of developing a red flowered Marvel-of-Peru plant**

	R	r
R	RR	Rr
r	Rr	rr

Try A 25% B 50% C 75% D 100%

Part 9 | Zama Try It 9.11 | PDF page 99 | Printed page 235

9.11 **The genotype of blood group O is**

Try It? A I^AI^B B I^Bi C ii D I^Ai

Part 9 | Zama Try It 9.12 | PDF page 99 | Printed page 235

9.12 **Blood group B has all of the following genotypes except**

Try It? A I^Bi B iI^B C I^BI^B D ii

Part 9 | Zama Try It 9.13 | PDF page 100 | Printed page 236

9.13 **If gametes have 23 chromosomes, how many pairs of chromosomes would you expect to see in kidney cells**

Try It? A 23 B 13 C 12.5 D 46

Part 9 | Zama Try It 9.14 | PDF page 100 | Printed page 236

9.14 **The number of autosomes present within a somatic cell in humans is**

Try A 41 B 42 C 43 D 44

Part 9 | Zama Try It 9.15 | PDF page 101 | Printed page 237

9.15 Telomeres are made up of

Try It?

A RNA + Proteins
B mRNA + Ribose
C DNA + Proteins
D tRNA + Proteins

Part 9 | Zama Try It 9.16 | PDF page 101 | Printed page 237

9.16 Down Syndrome is considered a

Try It?

A Trisomy
B Disomy
C Monosomy
D None of the above

Part 9 | Zama Try It 9.17 | PDF page 102 | Printed page 238

9.17 Which of the following genotypes is that of a normal male

Try It?

A XX B XO C XY D XXY

Part 9 | Zama Try It 9.18 | PDF page 103 | Printed page 239

9.18 Which of the following nitrogenous bases is seen in DNA but not RNA

Try It?

A Adenine
B Thymine
C Guanine
D Cytosine

Part 9 | Zama Try It 9.19 | PDF page 103 | Printed page 239

9.19 If Thymine constitutes 15% of DNA, what if the percentage of Guanine present

Try It?

A 35% B 45% C 25% D 15%

Part 9 | Zama Try It 9.20 | PDF page 104 | Printed page 240

9.20 A codon is made up of how many nucleotides

Try It? A 2 B 3 C 4 D 6

Part 9 | Zama Try It 9.21 | PDF page 104 | Printed page 240

9.21 Which of the following molecules is formed within the nucleolus, & binds with proteins in order to form ribosomes

Try It?
A Transfer RNA
B Messenger RNA
C Ribosomal RNA
D DNA Polymerase

Part 9 | Zama Try It 9.22 | PDF page 106 | Printed page 242

9.22 Which of the following sequences shows an insertion mutation in comparison to the original strand 5' GGGCCCAAA 3'

Try It?
A 5' GGGCCCAAA 3'
B 5' GGGAAACCC 3'
C 5' GGGCCAAA 3'
D 5' GGGCCCAAAA 3'

Part 9 | Zama Try It 9.23 | PDF page 106 | Printed page 242

9.23 A DNA strand has the following sequence, CCCCGAATT. A few years later, it was found that the same strand had changed into CCTCGAATT. What type of mutation is this?

Try It?
A Insertion Mutation
B Deletion Mutation
C Point Mutation
D None of the Above

Abdulkarim / Questions

Part 9 | Abdulkarim Q01 | PDF page 56

01/9

➢ **First scientist to study inheritance is.**

a) Mendel
b) Griffith
c) Bennet
d) Watson

Part 9 | Abdulkarim Q00 | PDF page 56

00/0

➢ **Mendel conducted his experiments on which of the following plants.**

a) Corn
b) Beans
c) Wheat
d) Pea

Part 9 | Abdulkarim Q03 | PDF page 56

03/9

➢ **According to the law of segregation, the ratio betwee offspring of the second generation is.**

a) 1 dominant : 1 recessive
b) 3 dominant : 1 recessive
c) 3 recessive : 1 dominant
d) 0 dominant : 1 recessive

Part 9 | Abdulkarim Q04 | PDF page 56

04/9

➢ **Crossing the green pea, yy, with the yellow pea, YY, th first generation results are…**

a) YY
b) yy
c) Yy
d) YY yy

Part 9 | Abdulkarim Q06 | PDF page 56

06/9

➢ **Pairs of alleles in a living organism are called**

a) appearance
b) Phenotype
c) external
d) Genotype

Part 9 | Abdulkarim Q09 | PDF page 57

09/9

➢ **An organism has 4 pairs of chromosomes, what is the possible combination of genes for this organism.**

a) 8
b) 16
c) 28
d) 32

Part 9 | Abdulkarim Q00 | PDF page 57

00/0

➢ **The possible combination of genes can be calculated by using which equation?**

a) n^4
b) 2^n
c) 4^n
d) n^2

Part 9 | Abdulkarim Q12 | PDF page 57

12/9

➢ **A recessive disease afflicting membrane protein.**

a) Galactosemia
b) Tay-Sachs disease (TSD)
c) Albinism
d) Cystic fibrosis

Part 9 | Abdulkarim Q13 | PDF page 57

$\frac{13}{9}$

A genetic disorder which results from the absence of th melanin pigment in both skin and hair.

a) Cystic fibrosis
b) Albinism
c) Tay-Sachs disease (TSD)
d) Galactosemia

Part 9 | Abdulkarim Q16 | PDF page 57

$\frac{16}{9}$

A heritable, genetic disorder resulting in the body's inability to digest galactose.

a) Cystic fibrosis
b) Galactosemia
c) Tay-Sachs disease (TSD)
d) Albinism

Part 9 | Abdulkarim Q17 | PDF page 57

$\frac{17}{9}$

An individual who is heterozygous for a recessive disorder

a) Disease carrier
b) Pedigree carrier
c) Trait carrier
d) Gene carrier

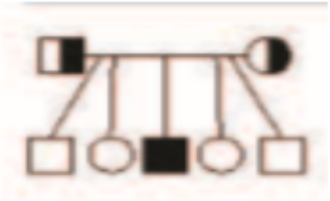
Part 9 | Abdulkarim Q18 | PDF page 58

$\frac{18}{9}$

Huntington's disease affects which system?

a) Nervous system
b) Reproductive system
c) Digestive system
d) Respiratory system

Part 9 | Abdulkarim Q19 | PDF page 58



$\frac{19}{9}$

The following diagram represents a pedigree for two parents and their offspring, to show the inheritance of Huntington's disease. from the diagram we can concluc that…

a) The father is normal

b) One of the offspring is normal

c) All the offspring are affected

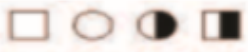
d) One of the offspring is affected

Part 9 | Abdulkarim Q21 | PDF page 58



$\frac{21}{9}$

The following diagram is a pedigree for certain traits exhibited by the parents, which of the following choice represents the offspring's genotype?

a) 

b) 

c) 

d) 

Part 9 | Abdulkarim Q22 | PDF page 58

22/9 ➢ **Sickle-cell disease follows the inheritance of**

a) Complete dominance
b) Incomplete dominance
c) Codominance
d) Mendelian dominance

Part 9 | Abdulkarim Q24 | PDF page 59

24/9 ➢ **In a hospital, four families disputed over their newbon's kinship, if the newborn's blood type is O, then which family cannot be related to the newborn?**

a) Father A, mother B
b) Father AB, mother O
c) Father B, mother 0
d) Father O, mother A

Part 9 | Abdulkarim Q25 | PDF page 59

25/9 ➢ **The blood type I^A and I^B is an example of.**

a) Complete dominance
b) Codominance
c) Incomplete dominance
d) Mendelian dominance

Part 9 | Abdulkarim Q26 | PDF page 60

$\frac{26}{9}$

➢ **The fur color of rabbits follow which inheritance ?**

a) Multiple alleles
b) Dominant lethal alleles
c) Recessive lethal alleles
d) Sex-linked traits

Part 9 | Abdulkarim Q27 | PDF page 60

$\frac{27}{9}$

➢ **What is the phenotype of the following figure**

a) CC
b) $c^{ch}c$
c) $c^{h}c^{h}$
d) cc

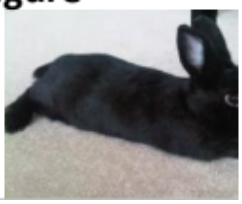


Part 9 | Abdulkarim Q28 | PDF page 60

$\frac{28}{9}$

➢ **What is the phenotype of the following figure**

a) CC
b) $c^{ch}c$
c) $c^{h}c^{h}$
d) cc



Part 9 | Abdulkarim Q30 | PDF page 60

$\frac{30}{9}$

➢ **The number of chromosomes in human's liver cells is**

a) 23
b) 44
c) 46
d) 69

Part 9 | Abdulkarim Q32 | PDF page 60

$\frac{32}{9}$

➢ **What determines the gender in humans?**

a) Chromosome number 21
b) Chromosomes x and y
c) codominance
d) Gene dominance

Part 9 | Abdulkarim Q33 | PDF page 60

$\frac{33}{9}$

➢ **Where are Barr bodies found?**

a) Female body cells
b) Female sex cells
c) Male body cells
d) Male sex cells

Part 9 | Abdulkarim Q34 | PDF page 61

$\frac{34}{9}$

➢ **Traits controlled by genes carried on an X chromosome …**

a) Sex Linked traits
b) Sex Influenced traits
c) Dominant Lethal genes
d) Recessive lethal genes

Part 9 | Abdulkarim Q35 | PDF page 61

35/9

➢ **A disease related to chromosomes responsible for determining the gender of the baby**

a) Short sightedness
b) Down's syndrome
c) Albinism
d) Hemophilia

Part 9 | Abdulkarim Q36 | PDF page 61

36/9

➢ **Father diagnosed with color blindness has a healthy daughter who married a healthy man: what's the percentage, their children will be diagnosed with color blindness**

a) 0%
b) 50%
c) 25%
d) 100%

Part 9 | Abdulkarim Q37 | PDF page 61

37/9

➢ **Which of the following is influenced by sex?**

e) Baldness
f) Color blindness
g) hemophilia
h) Albinism

Part 9 | Abdulkarim Q38 | PDF page 61

38/9

➢ **Baldness is affected by sex, dominant in males and recessive in females, if B represents "Baldness" and b represents "non-bald" then which of the following represents bald female genes?**

a) bb
b) bB
c) Bb
d) BB

Part 9 | Abdulkarim Q39 | PDF page 61

39/9

➢ **Which of the following genotypes gives a skin color similar to that of an AABBcc genotype?**

a) AaBbCc
b) aaBBcc
c) AABbCC
d) AaBBCc

Part 9 | Abdulkarim Q40 | PDF page 61

40/9

➢ **Which of the following phrases is not true regarding telomeres?**

a) Found in the end of a chromosome
b) Formed by DNA and sugars
c) Protects the chromosome
d) Plays a role in aging

Part 9 | Abdulkarim Q41 | PDF page 61

41/9

➢ **While making a karyotype for a baby, it was found that they had 3 copies of chromosome containing number 21, this baby is diagnosed with ...**

a) Turner's syndrome
b) Klinefelter's syndrome
c) Down's syndrome
d) Barre's syndrome

Part 9 | Abdulkarim Q43 | PDF page 62

43/9

➢ **Which of the following is the genotype for Klinefelter syndrome?**

a) OY
b) XO
c) XXY
d) XYY

Part 9 | Abdulkarim Q44 | PDF page 62

$\frac{44}{9}$ ➢ **Which of the following genotypes causes death?**

a) OY b) XO
c) XXY d) XYY

Part 9 | Abdulkarim Q45 | PDF page 62

$\frac{45}{9}$ ➢ **The first who led to the discovery of DNA as the genetic material is:**

a) Avery b) Griffith
c) Hershey and Chase d) Chargaff

Part 9 | Abdulkarim Q46 | PDF page 62

$\frac{46}{9}$ ➢ **Analyzed the amounts of Adenine, Guanine, Thymine, and Cytosine in the DNA:**

a) Chargaff b) Watson
c) Hershey d) Chase

Part 9 | Abdulkarim Q47 | PDF page 62

$\frac{47}{9}$ ➢ **The subunits for both DNA and RNA…**

a) Ribose b) Purines
c) Nucleotides d) Phosphor

Part 9 | Abdulkarim Q48 | PDF page 62

$\frac{48}{9}$ ➢ **The acid that carries the genetic material**

a) Amino b) Nucleic
c) Fatty d) Glycolic

Part 9 | Abdulkarim Q49 | PDF page 62

$\frac{49}{9}$ ➢ **Nucleotides in RNA contain ……. sugar**

a) Glucose b) Maltose
c) Sucrose d) Ribose

Part 9 | Abdulkarim Q50 | PDF page 62

$\frac{50}{9}$ ➢ **The nitrogenous base that's not found on RNA**

a) Cytosine b) Uracil
c) Thymine d) Guanine

Part 9 | Abdulkarim Q51 | PDF page 63

$\frac{51}{9}$ ➢ **Which of the following nitrogenous bases isn't from pyrimidines?**

a) Thymine b) Cytosine
c) Guanine d) Uracil

Part 9 | Abdulkarim Q52 | PDF page 63

52/9

➢ **Which of the following is the correct bonding between nitrogenous bases?**

a) A – T / C – G

b) G – T / A – C

c) A – G / C – T

d) U – C / A – G

Part 9 | Abdulkarim Q53 | PDF page 63

53/9

➢ **If the percentage of Thymine in DNA is 29%, then what the percentage of Adenine?**

a) 58%

b) 29%

c) 21%

d) 15%

Part 9 | Abdulkarim Q54 | PDF page 63

54/9

➢ **The enzyme responsible for unwinding the DNA during DNA replication is……**

a) RNA primase

b) DNA Ligase

c) DNA polymerase

d) DNA Helicase

Part 9 | Abdulkarim Q55 | PDF page 63

55/9

➢ **if the sequence of the nitrogenous bases in one of the DNA strands is: 5'CTGAATTCA3'; then what's the sequence of the complementary strand?**

a) 3’GACTTAAGT5’
b) 3’TCAGGCCTG5’
c) 3’AGTCCGGAT 5'
d) 3’CAGTTAACG 5’

Part 9 | Abdulkarim Q56 | PDF page 63

56/9

➢ **Carries the genetic information from the DNA in the nucleus to start protein synthesis**

a) RNA primase
b) Messenger RNA (mRNA)
c) Ribosomal RNA (rRNA)
d) Transfer RNA(tRNA)

Part 9 | Abdulkarim Q57 | PDF page 63

57/9

➢ **Which of the following transfers amino acids to the ribosome?**

a) RNA primase
b) Messenger RNA (mRNA)
c) Ribosomal RNA (rRNA)
d) Transfer RNA(tRNA)

Part 9 | Abdulkarim Q58 | PDF page 63

$\frac{58}{9}$ ➢ **If we wanted to make a protein with 60 amino acids, then the number of nitrogenous bases on the nucleic acid mRNA is …**

a) 60 b) 120
c) 360 d) 180

Part 9 | Abdulkarim Q59 | PDF page 64

$\frac{59}{9}$ ➢ **If the pattern of the nitrogenous base in a DNA segment is ATCAATTGG; then the mRNA is**

a) UAGUUAACC b) TAGTTAACC
c) AUCAAUUGG d) ATCAATTGG

Part 9 | Abdulkarim Q60 | PDF page 64

$\frac{60}{9}$ ➢ **Which one is a starting codon?**

a) AUG b) AUU
a) CAU b) UAA

Part 9 | Abdulkarim Q61 | PDF page 64

$\frac{61}{9}$ ➢ **Which is a stopping codon in mRNA?**

a) AUG b) AUU
c) CAU d) UAA

Part 9 | Abdulkarim Q62 | PDF page 64

$\frac{62}{9}$ ➢ **A process that binds mRNA with the ribosome and protein synthesis:**

a) Transcription
b) Gene regulation
c) Translation
d) Insertion

Part 9 | Abdulkarim Q63 | PDF page 64

$\frac{63}{9}$ ➢ **A segment of the DNA holds the following sequence: CCCCGAATT is mutated then the new sequence happens to be CCTCGAATT, what type of mutation is it?**

a) Duplication
b) Substitution
c) Deletion
d) Insertion

Part 9 | Abdulkarim Q64 | PDF page 64

$\frac{64}{9}$ ➢ **A segment of the DNA has TTAGGACCC sequence, which of the following shows an insertion mutation?**

a) TTACGACCC
b) TTAGACCC
c) TTAGGACCCTCC
d) TTAGGACCC

Part 9 | Abdulkarim Q65 | PDF page 64

$\frac{65}{9}$ ➢ **A segment of the DNA has GGG sequence and became GGA, what type of mutation is it?**

a) Deletion
b) Substitution
c) Insertion
d) Frameshift

Part 9 | Abdulkarim Q66 | PDF page 64

66/9 ➤ **Which of the following <u>is not</u> type of mutation?**

a) Substituting a base
b) RNA interference
c) Insertion
d) Frameshift

Answer-key appendix

Answers are listed only when reliably matched to the English choices. A differing scanned-key letter is retained explicitly; it is not silently overwritten. Unresolved entries have no scored answer.

Part 1

Source occurrence	PDF page	Answer	Verification note
Zama Worked 1.1	4	C	Matched to source
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Zama Worked 1.3	4	A	Matched to source
Zama Worked 1.4	4	C	Matched to source
Zama Worked 1.5	5	A	Matched to source
Zama Worked 1.6	5	A	Matched to source
Zama Worked 1.7	5	Unresolved	Not graded; source wording or answer needs review
Zama Worked 1.8	5	D	Matched to source
Zama Worked 1.9	5	C	Matched to source
Zama Worked 1.10	5	B	Matched to source
Zama Worked 1.11	6	B	Matched to source
Zama Worked 1.12	6	Unresolved	Not graded; source wording or answer needs review
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Abdulkarim Q26	5	C	English choices checked
Abdulkarim Q27	5	D	English choices checked

Part 2

Source occurrence	PDF page	Answer	Verification note
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Abdulkarim Q06	7	A	English choices checked
Abdulkarim Q07	7	B	English choices checked
Abdulkarim Q08	8	B	English choices checked
Abdulkarim Q09	8	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q10	8	B	English choices checked
Abdulkarim Q11	8	D	English choices checked
Abdulkarim Q12	8	A	English choices checked
Abdulkarim Q13	8	B	English choices checked
Abdulkarim Q14	8	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q15	8	C	English choices checked
Abdulkarim Q16	8	B	English choices checked

Part 9

Source occurrence	PDF page	Answer	Verification note
Zama Worked 9.1	93	A	Matched to source
Zama Worked 9.2	93	C	Matched to source
Zama Worked 9.3	93	A	Matched to source
Zama Worked 9.4	93	D	Matched to source
Zama Worked 9.5	93	A	Matched to source
Zama Worked 9.6	94	A	Matched to source
Zama Worked 9.7	94	D	Matched to source
Zama Worked 9.8	95	D	Matched to source
Zama Worked 9.9	95	C	Matched to source
Zama Worked 9.10	95	Unresolved	Not graded; source wording or answer needs review
Zama Worked 9.11	96	A	Matched to source
Zama Worked 9.12	96	A	Matched to source
Zama Worked 9.13	96	B	Matched to source
Zama Worked 9.14	96	D	Matched to source
Zama Worked 9.15	96	D	Matched to source
Zama Worked 9.16	96	A	Matched to source
Zama Worked 9.17	96	C	Matched to source
Zama Worked 9.18	96	A	Matched to source
Zama Worked 9.19	97	A	Matched to source
Zama Worked 9.20	97	B	Matched to source
Zama Worked 9.21	97	D	Matched to source
Zama Worked 9.22	97	C	Matched to source
Zama Worked 9.23	97	B	Matched to source
Zama Worked 9.24	98	B	Matched to source
Zama Worked 9.25	98	C	Matched to source
Zama Worked 9.26	98	A	Matched to source

Source occurrence	PDF page	Answer	Verification note
Zama Worked 9.27	98	B	Matched to source
Zama Worked 9.28	99	B	Matched to source
Zama Worked 9.29	99	D	Matched to source
Zama Worked 9.30	99	B	Matched to source
Zama Worked 9.31	99	B	Matched to source
Zama Worked 9.32	99	B	Matched to source
Zama Worked 9.33	99	D	Matched to source
Zama Worked 9.34	100	C	Matched to source
Zama Worked 9.35	100	C	Matched to source
Zama Worked 9.36	101	B	Matched to source
Zama Worked 9.37	101	D	Matched to source
Zama Worked 9.38	101	A	Matched to source
Zama Worked 9.39	101	Unresolved	Not graded; source wording or answer needs review
Zama Worked 9.40	101	D	Matched to source
Zama Worked 9.41	101	D	Matched to source
Zama Worked 9.42	101	Unresolved	Not graded; source wording or answer needs review
Zama Worked 9.43	101	B	Matched to source
Zama Worked 9.44	102	A	Matched to source
Zama Worked 9.45	102	B	Matched to source
Zama Worked 9.46	102	A	Matched to source
Zama Worked 9.47	102	A	Matched to source
Zama Worked 9.48	102	Unresolved	Not graded; source wording or answer needs review
Zama Worked 9.49	102	A	Matched to source
Zama Worked 9.50	102	C	Matched to source
Zama Worked 9.51	103	A	Matched to source
Zama Worked 9.52	103	D	Matched to source

Source occurrence	PDF page	Answer	Verification note
Zama Worked 9.53	103	A	Matched to source
Zama Worked 9.54	103	C	Matched to source
Zama Worked 9.55	103	C	Matched to source
Zama Worked 9.56	104	A	Matched to source
Zama Worked 9.57	104	D	Matched to source
Zama Worked 9.58	104	C	Matched to source
Zama Worked 9.59	104	A	Matched to source
Zama Worked 9.60	104	C	Matched to source
Zama Worked 9.61	105	A	Matched to source
Zama Worked 9.62	105	D	Matched to source
Zama Worked 9.63	105	C	Matched to source
Zama Worked 9.64	105	B	Matched to source
Zama Worked 9.65	106	B	Matched to source
Zama Worked 9.66	106	Unresolved	Not graded; source wording or answer needs review
Zama Worked 9.67	106	C	Matched to source
Zama Try 9.1	94	A	Matched to source
Zama Try 9.2	94	D	Matched to source
Zama Try 9.3	94	C	Matched to source
Zama Try 9.4	95	Unresolved	Not graded; source wording or answer needs review
Zama Try 9.5	96	C	Matched to source
Zama Try 9.6	96	Unresolved	Not graded; source wording or answer needs review
Zama Try 9.7	96	Unresolved	Not graded; source wording or answer needs review
Zama Try 9.8	97	C	Matched to source
Zama Try 9.9	98	C	Matched to source
Zama Try 9.10	98	A	Matched to source
Zama Try 9.11	99	C	Matched to source

Source occurrence	PDF page	Answer	Verification note
Zama Try 9.12	99	D	Matched to source
Zama Try 9.13	100	A	Matched to source
Zama Try 9.14	100	D	Matched to source
Zama Try 9.15	101	C	Matched to source
Zama Try 9.16	101	A	Matched to source
Zama Try 9.17	102	C	Matched to source
Zama Try 9.18	103	B	Matched to source
Zama Try 9.19	103	A	Matched to source
Zama Try 9.20	104	B	Matched to source
Zama Try 9.21	104	C	Matched to source
Zama Try 9.22	106	D	Matched to source
Zama Try 9.23	106	C	Matched to source
Abdulkarim Q01	56	A	English choices checked
Abdulkarim Q00	56	D	English choices checked
Abdulkarim Q03	56	B	English choices checked
Abdulkarim Q04	56	C	English choices checked
Abdulkarim Q06	56	D	Scanned key: C; English-choice solution: D
Abdulkarim Q09	57	B	Scanned key: C; English-choice solution: B
Abdulkarim Q00	57	B	English choices checked
Abdulkarim Q12	57	D	English choices checked
Abdulkarim Q13	57	B	English choices checked
Abdulkarim Q16	57	B	English choices checked
Abdulkarim Q17	57	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q18	58	A	English choices checked
Abdulkarim Q19	58	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q21	58	A	English choices checked

Source occurrence	PDF page	Answer	Verification note
Abdulkarim Q22	58	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q24	59	B	English choices checked
Abdulkarim Q25	59	B	English choices checked
Abdulkarim Q26	60	A	English choices checked
Abdulkarim Q27	60	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q28	60	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q30	60	C	Scanned key: B; English-choice solution: C
Abdulkarim Q32	60	B	English choices checked
Abdulkarim Q33	60	A	English choices checked
Abdulkarim Q34	61	A	English choices checked
Abdulkarim Q35	61	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q36	61	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q37	61	A	English choices checked
Abdulkarim Q38	61	D	English choices checked
Abdulkarim Q39	61	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q40	61	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q41	61	C	English choices checked
Abdulkarim Q43	62	C	English choices checked
Abdulkarim Q44	62	A	English choices checked
Abdulkarim Q45	62	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q46	62	A	English choices checked
Abdulkarim Q47	62	C	English choices checked
Abdulkarim Q48	62	B	English choices checked
Abdulkarim Q49	62	D	English choices checked
Abdulkarim Q50	62	C	English choices checked
Abdulkarim Q51	63	C	English choices checked

Source occurrence	PDF page	Answer	Verification note
Abdulkarim Q52	63	A	English choices checked
Abdulkarim Q53	63	B	English choices checked
Abdulkarim Q54	63	D	English choices checked
Abdulkarim Q55	63	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q56	63	B	English choices checked
Abdulkarim Q57	63	D	English choices checked
Abdulkarim Q58	63	D	Scanned key: C; English-choice solution: D
Abdulkarim Q59	64	A	English choices checked
Abdulkarim Q60	64	Unresolved	Not graded; source wording or answer needs review
Abdulkarim Q61	64	D	English choices checked
Abdulkarim Q62	64	C	English choices checked
Abdulkarim Q63	64	B	English choices checked
Abdulkarim Q64	64	C	English choices checked
Abdulkarim Q65	64	B	English choices checked
Abdulkarim Q66	64	B	English choices checked